

A WKB-based fixed-grid method for capturing trait concentration in a dispersal evolution model

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Abstract

The evolution of dispersal traits is a fundamental topic in evolutionary ecology, where natural selection may drive the trait distribution toward concentration in the rare-mutation regime. This singular behavior poses a serious numerical difficulty, since direct discretizations of the population density require very fine trait grids to identify the fittest trait and to resolve the concentrated profile accurately. In this paper, we develop a WKB-based numerical framework for a dispersal evolution model. By separating the exponentially concentrated trait dependence from a smoother amplitude variable and combining this WKB representation with dual trait-grid implementation and other specially designed techniques, the method recovers the selected trait and the associated concentration structure accurately and efficiently on fixed trait grids. We establish a semi-discrete stationary fixed-grid asymptotic-preserving structure for the rare-mutation limit of the steady-state problem. Numerical experiments compare the proposed method with direct density discretizations and confirm its advantage in the small-mutation regime.

Key words. dispersal evolution, rare-mutation limit, trait concentration, WKB reformulation, Hamilton–Jacobi limit, asymptotic-preserving structure.

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1 Introduction

Adaptive dynamics studies the evolution of phenotypic traits under mutation, selection, and ecological feedback. In structured population models, natural selection is often manifested by the concentration of the population density on one or several fittest traits. In the rare-mutation regime, such concentration is usually described by a measure-valued limit in the trait variable, and the support of the limiting measure represents the selected phenotype. This viewpoint has been developed in selection-mutation models and in the Hamilton–Jacobi approach to adaptive dynamics [3, 8]. Among these problems, the evolution of dispersal is a particularly important example. In spatially heterogeneous environments, different dispersal rates may lead to different fitness advantages, and the selected strategy depends on the interaction between movement, resource distribution, and population competition. This question has been studied from several viewpoints in mathematical ecology, for example in reaction-diffusion models for slow dispersal, ideal free distributions, and spatial sorting [9, 11, 23].

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In this work, we consider a representative dispersal evolution model of the form

$$\varepsilon \partial_t n_\varepsilon - D(\theta) \Delta_{\mathbf{x}} n_\varepsilon - \varepsilon^2 \partial_{\theta\theta} n_\varepsilon = n_\varepsilon (K(\mathbf{x}) - \rho_\varepsilon(t, \mathbf{x})), \quad (1.1)$$

where $\mathbf{x} \in \Omega \subset \mathbb{R}^d$, $d = 1, 2, 3$, $\theta \in \mathbb{T} = (0, 1]$. Here t denotes time, $\mathbf{x}$ is the spatial variable, and θ is the phenotypic trait. The unknown $n_\varepsilon(t, \mathbf{x}, \theta)$ is the structured population density, and

$$\rho_\varepsilon(t, \mathbf{x}) = \int_{\mathbb{T}} n_\varepsilon(t, \mathbf{x}, \theta) d\theta, \quad P_\varepsilon(t, \theta) = \int_{\Omega} n_\varepsilon(t, \mathbf{x}, \theta) d\mathbf{x}, \quad (1.2)$$

denote the population density in space and the trait marginal, respectively. The concentration of P_ε in the rare-mutation regime is used to identify the selected trait. The function $K(\mathbf{x})$ represents the local carrying capacity, while the trait-dependent diffusivity $D(\theta)$ models the dispersal rate associated with phenotype θ . Throughout the paper, we assume that the dispersal rate is positive and periodic in the trait variable and that the carrying capacity is nonnegative and bounded [22]

$$0 < D_{\min} \leq D(\theta) \leq D_{\max} < \infty, \quad 0 \leq K(\mathbf{x}) \leq K_{\max}. \quad (1.3)$$

We impose homogeneous Neumann boundary conditions on $\partial\Omega$ and periodic boundary conditions in the trait direction. The time-dependent model (1.1) is used here as an evolutionary relaxation toward the selection state, while the asymptotic structure of interest is motivated by the rare-mutation limit of the associated steady-state problem.

The steady-state version of this dispersal evolution model was studied by B. Perthame and P. E. Souganidis in [22]. Their work provides a rare-mutation description of the selected dispersal trait for a population structured by space and by a continuous dispersal trait. In the limit of vanishing mutations, the population concentrates in the trait variable, while the spatial profile remains nontrivial. This result gives the asymptotic background for the present paper and clarifies the limiting selection structure that the numerical method aims to capture.

This dispersal model fits into a broader class of structured selection models with concentration in the small-mutation limit. For purely trait-structured or nonlocal selection models, Dirac concentration and constrained Hamilton–Jacobi equations have been studied in [7] and [18]. When space is included as an additional structuring variable, the analysis becomes more delicate. Related Hamilton–Jacobi limits for populations structured by space and trait were obtained in [6], and an asymptotic analysis of a selection model with space was given in [20]. The Perthame–Souganidis dispersal model has also motivated later studies on dispersal evolution, conditional dispersal, and the stability of Dirac concentrations [10, 16, 17].

The same concentration mechanism creates a severe numerical difficulty. Direct discretizations of the original density must resolve a trait profile whose width shrinks with the mutation scale. Hence the mesh required in the trait direction may become prohibitively fine when ε is small. This suggests that the numerical method should exploit the asymptotic separation between the singular trait dependence and the smoother amplitude, rather than resolve the concentrated density directly.

This is closely related to the idea of asymptotic-preserving schemes, whose goal is to capture the limiting regime on discretizations that do not resolve the small scale ε [14]. In rare-mutation problems, the natural asymptotic separation is provided by a Wentzel–Kramers–Brillouin (WKB) [5] or Hopf–Cole transformation [4]. The density is represented by an exponential phase, which describes the concentration in the trait variable, multiplied by a smoother amplitude. Such a WKB viewpoint is standard in the Hamilton–Jacobi approach to adaptive dynamics [3] and is also central in the dispersal model of [22].

WKB-type transformations have been used as numerical tools in different multiscale regimes. For highly oscillatory problems, such as semiclassical Schrödinger equations, the exponential phase

is typically imaginary or complex-valued, and the WKB transformation separates the rapid oscillations from smoother amplitude variables [2, 15]. In selection-mutation problems, by contrast, the phase is real-valued and the small parameter leads to concentration rather than oscillation. In adaptive dynamics, a WKB representation has been used to construct an asymptotic-preserving scheme for capturing Dirac concentrations on coarse, ε -independent phenotype meshes [1]. Related Hopf–Cole based ideas have been developed for kinetic reaction–transport equations in singular front propagation regimes [12], and for kinetic BGK models in the hyperbolic limit [19]. These works motivate the WKB-based formulation adopted here for the dispersal evolution model.

The present paper develops a WKB-based fixed-grid numerical framework for the dispersal evolution problem (1.1). Starting from the phase-amplitude representation, we derive a semi-discrete scheme for the phase and amplitude variables and prove the positivity of the amplitude. We then establish a stationary fixed-grid asymptotic-preserving structure associated with the rare-mutation limit of the steady-state problem under a one-sided remainder stability condition. For computation, we propose a dual trait-grid implementation. The one-dimensional phase is resolved on a fine trait grid, whereas the space-dependent amplitude is evolved on a coarser trait grid. This separation preserves the main advantage of the WKB formulation and reduces the dominant cost associated with the space-dependent unknown.

The paper is organized as follows. Section 2 recalls the WKB ansatz and the steady-state rare-mutation structure that motivates the numerical formulation. Section 3 presents the semi-discrete WKB scheme, proves positivity of the amplitude, and establishes the stationary fixed-grid asymptotic-preserving structure. Section 4 describes the fully discrete implementation used in the computations and presents numerical experiments comparing the WKB method with direct density discretizations in the small-mutation regime. The appendices collect the auxiliary numerical Hamiltonians, fluxes, and remainder estimates used in the analysis. Finally, conclusions are drawn in Section 5.

2 Preliminaries

This section summarizes the continuous asymptotic structure that motivates the WKB reformulation and the subsequent numerical discretization.

2.1 WKB ansatz and reformulated system

For simplicity, we restrict the presentation to the one-dimensional spatial case, i.e. $x \in \Omega \subset \mathbb{R}$. To capture the concentration phenomenon in the trait variable, we introduce the classical WKB ansatz

$$n_\varepsilon(t, x, \theta) = W_\varepsilon(t, x, \theta) \exp\left(\frac{u_\varepsilon(t, \theta)}{\varepsilon}\right). \quad (2.1)$$

This decomposition separates the exponentially varying part in the trait direction from a comparatively smoother amplitude. It is therefore natural for both the continuous asymptotic analysis and the numerical design in the small-mutation regime.

Substituting (2.1) into (1.1) and assigning the leading trait-direction exponential contributions to the phase function, we obtain the reformulated system

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon \partial_{\theta\theta} u_\varepsilon = -\mathcal{H}, \\ \varepsilon \partial_t W_\varepsilon - D(\theta) \Delta_x W_\varepsilon - \varepsilon^2 \partial_{\theta\theta} W_\varepsilon - 2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon = W_\varepsilon (K(x) - \rho_\varepsilon(t, x) + \mathcal{H}). \end{cases} \quad (2.2)$$

Here $\mathcal{H}$ is introduced as an effective Hamiltonian. As we can see later, a natural choice is given by the principal eigenvalue of an associated spatial eigenvalue problem (2.9). Using the identity

$$\partial_\theta(W_\varepsilon \partial_\theta u_\varepsilon) = \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon + W_\varepsilon \partial_{\theta\theta} u_\varepsilon,$$

we rewrite the term $-2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon$ in (2.2) as $2\varepsilon \partial_\theta F_\varepsilon + 2\varepsilon W_\varepsilon \partial_{\theta\theta} u_\varepsilon$, where $F_\varepsilon = -W_\varepsilon \partial_\theta u_\varepsilon$. Consequently, the system (2.2) can also be written equivalently as

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon \partial_{\theta\theta} u_\varepsilon = -\mathcal{H}, \\ \varepsilon \partial_t W_\varepsilon - D(\theta) \Delta_x W_\varepsilon - \varepsilon^2 \partial_{\theta\theta} W_\varepsilon + 2\varepsilon \partial_\theta F_\varepsilon = W_\varepsilon (K(x) - \rho_\varepsilon(t, x) + \mathcal{H} - 2\varepsilon \partial_{\theta\theta} u_\varepsilon). \end{cases} \quad (2.3)$$

2.2 Continuous asymptotic structure of the steady state

We next recall the steady-state asymptotic structure that motivates the numerical method. Let $(\bar{n}_\varepsilon, \bar{\rho}_\varepsilon)$ be a positive steady state of (1.1). Then

$$-D(\theta) \Delta_x \bar{n}_\varepsilon - \varepsilon^2 \partial_{\theta\theta} \bar{n}_\varepsilon = \bar{n}_\varepsilon (K(x) - \bar{\rho}_\varepsilon(x)), \quad \bar{\rho}_\varepsilon(x) = \int_{\mathbb{T}} \bar{n}_\varepsilon(x, \theta) d\theta. \quad (2.4)$$

As shown in [22], under standard assumptions on K , D , and the domain, the steady-state family admits several properties that are fundamental for the rare-mutation limit.

Theorem 2.1 (Continuous asymptotic structure of the steady state). *Assume that K is positive and nonconstant, that D is positive, periodic, and has a unique minimizer θ_m , and that the steady problem (2.4) admits a positive solution. Then the following properties hold at the continuous level.*

1. *There exists a constant $C > 0$, independent of ε , such that*

$$0 \leq \bar{\rho}_\varepsilon(x) \leq C \quad \text{for all } x \in \Omega. \quad (2.5)$$

Moreover, up to subsequences, $\bar{\rho}_\varepsilon$ converges uniformly to a limit density ρ .

2. *If we write*

$$\bar{n}_\varepsilon(x, \theta) = \exp\left(\frac{\bar{u}_\varepsilon(x, \theta)}{\varepsilon}\right), \quad (2.6)$$

then $\bar{u}_\varepsilon$ is uniformly Lipschitz continuous in θ , and along subsequences

$$\bar{u}_\varepsilon \rightarrow u \quad \text{uniformly in } (x, \theta), \quad (2.7)$$

where the limit u is independent of x , periodic in θ , and satisfies

$$\begin{cases} |\partial_\theta u(\theta)|^2 = \mathcal{H}(\theta, \rho(\cdot)), \\ \max_{\theta \in \mathbb{T}} u(\theta) = 0. \end{cases} \quad (2.8)$$

3. *The effective Hamiltonian $\mathcal{H}(\theta, \rho)$ is defined through the principal eigenvalue problem*

$$\begin{cases} -D(\theta) \Delta_x \mathcal{N}(x, \theta) = \mathcal{N}(x, \theta) (K(x) - \rho(x)) + \mathcal{N}(x, \theta) \mathcal{H}(\theta, \rho(\cdot)), & x \in \Omega, \\ \partial_\nu \mathcal{N} = 0, & x \in \partial\Omega, \end{cases} \quad (2.9)$$

with a positive eigenfunction $\mathcal{N}$, where ν denotes the outward unit normal vector on $\partial\Omega$. Moreover, $\mathcal{H}$ has the same monotonicity in θ as D , and hence

$$\min_{\theta \in \mathbb{T}} \mathcal{H}(\theta, \rho(\cdot)) = \mathcal{H}(\theta_m, \rho(\cdot)) = 0. \quad (2.10)$$

4. The steady state concentrates on the fittest trait in the limit $\varepsilon \rightarrow 0$. More precisely,

$$\bar{n}_\varepsilon \rightarrow N_m(x) \delta(\theta - \theta_m), \quad \bar{\rho}_\varepsilon \rightarrow N_m(x), \quad (2.11)$$

where N_m solves the reduced Fisher-type problem

$$\begin{cases} -D(\theta_m) \Delta_x N_m = N_m (K(x) - N_m), & x \in \Omega, \\ \partial_\nu N_m = 0, & x \in \partial\Omega. \end{cases} \quad (2.12)$$

Theorem 2.1 recalls the continuous rare-mutation structure that motivates the numerical construction. It shows that the selected trait is encoded by the limiting phase, while the spatial profile is determined through an effective Hamiltonian. The discrete counterpart of this structure will be discussed later in Section 3.

3 WKB reformulated numerical scheme

In this section, we develop and analyze a semi-discrete numerical scheme for the WKB reformulation (2.3). The scheme is formulated in terms of the phase variable u_ε and the amplitude variable W_ε , with the total density n_ε reconstructed from the WKB representation.

3.1 Semi-discrete formulation

We work at the semi-discrete level in order to separate the discretization in (x, θ) from the additional issue of time stepping. For clarity, we present the scheme in one spatial dimension with uniform grids in the x - and θ -directions. This choice is not essential for the WKB decomposition, but it keeps the notation transparent.

Let $\{x_j\}_{j=0}^{N_x}$ be a uniform grid on $\Omega = (a, b)$ and let $\{\theta_k\}_{k=1}^{N_\theta}$ be a uniform periodic grid on $\mathbb{T} = (0, 1]$, with mesh size Δx and $\Delta\theta$. We denote

$$W_{j,k}^\varepsilon(t) \approx W_\varepsilon(t, x_j, \theta_k), \quad u_k^\varepsilon(t) \approx u_\varepsilon(t, \theta_k).$$

For notational convenience, we write

$$D_k = D(\theta_k), \quad k = 1, 2, \dots, N_\theta, \quad K_j = K(x_j), \quad j = 0, 1, \dots, N_x.$$

By (1.3), we have

$$0 < D_{\min} \leq D_k \leq D_{\max} < \infty, \quad 0 \leq K_j \leq K_{\max}.$$

For a grid function $V = \{V_{j,k}\}$, we use the standard second-order difference operators

$$\delta_x^2 V_{j,k} = \frac{V_{j+1,k} - 2V_{j,k} + V_{j-1,k}}{(\Delta x)^2}, \quad \delta_\theta^2 V_{j,k} = \frac{V_{j,k+1} - 2V_{j,k} + V_{j,k-1}}{(\Delta\theta)^2}.$$

The homogeneous Neumann boundary condition in x is imposed by the ghost values

$$V_{-1,k} = V_{1,k}, \quad V_{N_x+1,k} = V_{N_x-1,k},$$

while all indices in the θ -direction are understood periodically. For the phase variable, we also use

$$\delta_\theta^- u_k = \frac{u_k - u_{k-1}}{\Delta\theta}, \quad \delta_\theta^+ u_k = \frac{u_{k+1} - u_k}{\Delta\theta}.$$

The semi-discrete WKB system involves two numerical ingredients that are not contained in the standard second-order difference operators introduced above. The first one is the numerical Hamiltonian for the phase equation. The Hamilton–Jacobi term $-|\partial_\theta u_\varepsilon|^2$ in (2.3) is approximated by

$$\widehat{H}(\delta_\theta^- u_k^\varepsilon, \delta_\theta^+ u_k^\varepsilon).$$

We assume that $\widehat{H}$ is locally Lipschitz and consistent with the continuous Hamiltonian, namely

$$\widehat{H}(p, p) = -p^2.$$

We also assume that it is monotone in the sense that it is nondecreasing with respect to its first argument and nonincreasing with respect to its second argument. When $\widehat{H}$ is differentiable, this condition reads

$$\partial_a \widehat{H}(a, b) \geq 0, \quad \partial_b \widehat{H}(a, b) \leq 0.$$

One typical example is the Godunov numerical Hamiltonian, shown in Appendix A.

The second ingredient is a conservative discretization of the transport coupling in the amplitude equation. Recall that the flux in (2.3) is $F_\varepsilon = -W_\varepsilon \partial_\theta u_\varepsilon$. Let $\widehat{\mathcal{F}}^\varepsilon$ be a numerical flux approximating F_ε . We denote by

$$\mathcal{D}_\theta \widehat{\mathcal{F}}_{j,k}^\varepsilon$$

a conservative and consistent approximation of $\partial_\theta F_\varepsilon$ at the grid point (x_j, θ_k) . A typical choice is the upwind flux described in Appendix A.

For the positivity argument below, we further assume that this discretization satisfies the quasi-positivity property. Specifically, for any fixed pair of indices (j, k) , if $W_{j,k} = 0$ and $W_{j,k'} \geq 0$ for all k' , then we have

$$-\mathcal{D}_\theta \widehat{\mathcal{F}}_{j,k}^\varepsilon \geq 0. \tag{3.1}$$

It is easy to check that the upwind flux described in Appendix A satisfies the property.

The semi-discrete WKB system. With these two ingredients specified, we now define the discrete density and the semi-discrete WKB system. For notational convenience, set

$$E_k^\varepsilon(t) = \exp\left(\frac{u_k^\varepsilon(t)}{\varepsilon}\right). \tag{3.2}$$

The reconstructed nodal density is defined by

$$n_{j,k}^\varepsilon(t) = W_{j,k}^\varepsilon(t) E_k^\varepsilon(t). \tag{3.3}$$

As we shall see later, the evolution of u_ε and W_ε is coupled through the spatial marginal density ρ_ε . In the semi-discrete scheme, this marginal is computed by the composite quadrature in the trait direction. More precisely, we define

$$\rho_j^\varepsilon(t) := \Delta\theta \sum_{k=1}^{N_\theta} n_{j,k}^\varepsilon(t) = \Delta\theta \sum_{k=1}^{N_\theta} W_{j,k}^\varepsilon(t) E_k^\varepsilon(t), \quad j = 0, 1, \dots, N_x, \tag{3.4}$$

which approximates $\rho_\varepsilon(t, x_j)$ in (1.2). Given

$$\boldsymbol{\rho}_\varepsilon(t) = (\rho_0^\varepsilon(t), \rho_1^\varepsilon(t), \dots, \rho_{N_x}^\varepsilon(t)),$$

the discrete effective Hamiltonian $\mathcal{H}_k(\boldsymbol{\rho}_\varepsilon(t))$, which approximates the principal eigenvalue $\mathcal{H}(\theta_k, \rho_\varepsilon(t))$ in (2.9), is determined by the discrete principal eigenvalue problem

$$\begin{cases} -D_k \delta_x^2 \mathcal{N}_j^{(k)} = \mathcal{N}_j^{(k)} (K_j - \rho_j^\varepsilon(t)) + \mathcal{N}_j^{(k)} \mathcal{H}_k(\boldsymbol{\rho}_\varepsilon(t)), & j = 0, 1, \dots, N_x, \\ \mathcal{N}_{-1}^{(k)} = \mathcal{N}_1^{(k)}, \quad \mathcal{N}_{N_x+1}^{(k)} = \mathcal{N}_{N_x-1}^{(k)}, \end{cases} \quad (3.5)$$

where $\mathcal{N}^{(k)}$ is the corresponding positive eigenvector.

With the above notation, the semi-discrete WKB scheme reads

$$\begin{cases} \frac{d}{dt} u_k^\varepsilon + \widehat{H}(\delta_\theta^- u_k^\varepsilon, \delta_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k(\boldsymbol{\rho}_\varepsilon(t)), & k = 1, \dots, N_\theta, \\ \varepsilon \frac{d}{dt} W_{j,k}^\varepsilon - D_k \delta_x^2 W_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon + 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}_{j,k} \\ = W_{j,k}^\varepsilon (K_j - \rho_j^\varepsilon(t) + \mathcal{H}_k(\boldsymbol{\rho}_\varepsilon(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon), & j = 0, 1, \dots, N_x, \quad k = 1, \dots, N_\theta. \end{cases} \quad (3.6)$$

The semi-discrete equation of n and the remainder $R_{j,k}^\varepsilon$. We next show the equation satisfied by the reconstructed density $n_{j,k}^\varepsilon$. Since E_k^ε is independent of the spatial index j ,

$$\delta_x^2 n_{j,k}^\varepsilon = E_k^\varepsilon \delta_x^2 W_{j,k}^\varepsilon.$$

Moreover,

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon = E_k^\varepsilon \left(\varepsilon \frac{d}{dt} W_{j,k}^\varepsilon + W_{j,k}^\varepsilon \frac{d}{dt} u_k^\varepsilon \right). \quad (3.7)$$

Multiplying the first equation in (3.6) by $E_k^\varepsilon W_{j,k}^\varepsilon$ and multiplying the second equation in (3.6) by $\varepsilon E_k^\varepsilon$, using (3.7), we can derive that $n_{j,k}^\varepsilon$ satisfies

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon = (K_j - \rho_j^\varepsilon) n_{j,k}^\varepsilon + R_{j,k}^\varepsilon, \quad (3.8)$$

where the remainder has the form

$$R_{j,k}^\varepsilon = \varepsilon^2 E_k^\varepsilon \delta_\theta^2 W_{j,k}^\varepsilon - \varepsilon W_{j,k}^\varepsilon E_k^\varepsilon \delta_\theta^2 u_k^\varepsilon - W_{j,k}^\varepsilon E_k^\varepsilon \widehat{H}(\delta_\theta^- u_k^\varepsilon, \delta_\theta^+ u_k^\varepsilon) - 2\varepsilon E_k^\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}_{j,k}^\varepsilon. \quad (3.9)$$

The equation (3.8) is a semi-discrete analogue of the original density equation (1.1), with the remainder $R_{j,k}^\varepsilon$ (3.9) which approximates $\varepsilon^2 \partial_{\theta\theta} n^\varepsilon$.

3.2 Basic structural property

The semi-discrete scheme (3.6) can be shown to be positive-preserving as long as the initial data is nonnegative. The detailed result is summarized as the following lemma.

Lemma 3.1 (Positivity of the amplitude). *Assume that $W_{j,k}^\varepsilon(0) \geq 0$ for all j, k . Then the semi-discrete amplitude equation in (3.6) preserves nonnegativity, namely*

$$W_{j,k}^\varepsilon(t) \geq 0 \quad \forall j, k, \quad t \geq 0,$$

as long as the solution exists. Consequently,

$$n_{j,k}^\varepsilon(t) \geq 0 \quad \forall j, k, \quad t \geq 0.$$

Proof. From the amplitude equation in (3.6), we can write

$$\varepsilon \frac{d}{dt} W_{j,k}^\varepsilon = D_k \delta_x^2 W_{j,k}^\varepsilon + \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon - 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}_{j,k}^\varepsilon + W_{j,k}^\varepsilon B_{j,k}(t),$$

where

$$B_{j,k}(t) = K_j - \rho_j^\varepsilon(t) + \mathcal{H}_k(\rho_\varepsilon(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon(t).$$

Let $W^\varepsilon \geq 0$ and suppose that $W_{j,k}^\varepsilon = 0$ for some (j, k) at time t . Then, for an interior point,

$$\delta_x^2 W_{j,k}^\varepsilon = \frac{W_{j+1,k}^\varepsilon + W_{j-1,k}^\varepsilon}{(\Delta x)^2} \geq 0, \quad \delta_\theta^2 W_{j,k}^\varepsilon = \frac{W_{j,k+1}^\varepsilon + W_{j,k-1}^\varepsilon}{(\Delta \theta)^2} \geq 0.$$

The same conclusion holds at the boundary because of the Neumann ghost values in x -direction and the periodicity in θ -direction. Combining all the inequalities, using the quasi-positivity assumption (3.1) and the fact that $W_{j,k}^\varepsilon B_{j,k}(t) = 0$ since we assumed $W_{j,k}^\varepsilon = 0$, we finally have that, at this specific (j, k) , the following inequality holds,

$$\left. \frac{d}{dt} W_{j,k}^\varepsilon \right|_{W_{j,k}^\varepsilon = 0} \geq 0.$$

Thus the vector field points inward on the boundary of the nonnegative cone. The standard invariance criterion for finite-dimensional ODE systems implies the claim. $\square$

3.3 Conditional stationary asymptotic-preserving structure

We now turn to stationary semi-discrete states. The goal is to identify which parts of the continuous rare-mutation structure are retained on a fixed grid. By Lemma 3.1, stationary states obtained from nonnegative amplitudes satisfy

$$W_{j,k}^\varepsilon \geq 0, \quad n_{j,k}^\varepsilon = W_{j,k}^\varepsilon E_k^\varepsilon \geq 0.$$

All estimates below are for such nonnegative reconstructed densities.

Weighted density balance. Define

$$\eta_j^\varepsilon = \Delta \theta \sum_{k=1}^{N_\theta} \frac{n_{j,k}^\varepsilon}{D_k}. \quad (3.10)$$

Recalling (3.4), multiplying (3.8) by $1/D_k$ and summing over the trait variable gives the stationary weighted balance equation

$$-\delta_x^2 \rho_j^\varepsilon = \eta_j^\varepsilon (K_j - \rho_j^\varepsilon) + \mathcal{R}_j^{\varepsilon,D}, \quad j = 0, 1, \dots, N_x, \quad (3.11)$$

where

$$\mathcal{R}_j^{\varepsilon,D} = \Delta \theta \sum_{k=1}^{N_\theta} \frac{R_{j,k}^\varepsilon}{D_k}. \quad (3.12)$$

The term $\mathcal{R}_j^{\varepsilon,D}$ is the discrete analogue of the continuous weighted mutation contribution

$$\varepsilon^2 \int_{\mathbb{T}} \frac{1}{D(\theta)} \partial_{\theta\theta} n^\varepsilon d\theta = \varepsilon^2 \int_{\mathbb{T}} n^\varepsilon \partial_{\theta\theta} \left(\frac{1}{D(\theta)} \right) d\theta, \quad (3.13)$$

where periodicity in the trait variable has been used. At the continuous level this term is bounded above by $C\varepsilon^2\rho^\varepsilon$. For the split WKB discretization, however, the discrete operators act on the reconstructed density $n_{j,k}^\varepsilon$ and an exact discrete chain rule is not available. Therefore, we present the following one-sided control assumption instead.

Assumption 3.1 (One-sided weighted remainder stability). *For fixed spatial and trait grids, there exist constants $C_{R,h} > 0$ and $r_h \geq 0$, possibly depending on the fixed mesh but independent of ε , such that every stationary state under consideration satisfies*

$$(\mathcal{R}_j^{\varepsilon,D})_+ \leq C_{R,h}\rho_j^\varepsilon + r_h, \quad j = 0, 1, \dots, N_x. \quad (3.14)$$

This condition controls only the positive part of the weighted defect. A negative contribution is favorable in the maximum-principle estimate. The assumption should be understood as a structural stability condition on the trait discretization. For a density-level conservative discretization, this type of estimate follows from periodic discrete summation by parts in the trait variable. Appendix B records this special case. For the WKB scheme used in the paper, the behavior of this estimate is examined numerically in Section 4. As we will see later, in the numerical tests, the weighted defect is typically nonpositive or has a small positive part, which supports the practical validity of the assumption.

Proposition 3.1 (Conditional density bound for stationary semi-discrete states). *Let $(u^\varepsilon, W^\varepsilon)$ be a stationary semi-discrete state with nonnegative reconstructed density. Assume that Assumption 3.1 holds. Then there exists a constant $C_\rho > 0$, independent of ε , such that*

$$0 \leq \rho_j^\varepsilon \leq C_\rho, \quad j = 0, 1, \dots, N_x.$$

Proof. By using $K_j \leq K_{\max}$ and Assumption 3.1, it is easy to get from (3.11) that

$$-\delta_x^2 \rho_j^\varepsilon + \eta_j^\varepsilon \rho_j^\varepsilon = K_j \eta_j^\varepsilon + \mathcal{R}_j^{\varepsilon,D} \leq K_{\max} \eta_j^\varepsilon + C_{R,h} \rho_j^\varepsilon + r_h.$$

Let j_* be a point where

$$\rho_{j_*}^\varepsilon = \max_j \rho_j^\varepsilon.$$

Since $\delta_x^2 \rho_{j_*}^\varepsilon \leq 0$, we have

$$\eta_{j_*}^\varepsilon \rho_{j_*}^\varepsilon \leq K_{\max} \eta_{j_*}^\varepsilon + C_{R,h} \rho_{j_*}^\varepsilon + r_h. \quad (3.15)$$

On the other hand, recalling (1.3) and (3.10), it is easy to see that, for all j ,

$$\frac{1}{D_{\max}} \rho_j^\varepsilon \leq \eta_j^\varepsilon \leq \frac{1}{D_{\min}} \rho_j^\varepsilon. \quad (3.16)$$

Combining (3.15) and (3.16) gives

$$\frac{(\rho_{j_*}^\varepsilon)^2}{D_{\max}} \leq \left(\frac{K_{\max}}{D_{\min}} + C_{R,h} \right) \rho_{j_*}^\varepsilon + r_h.$$

The right-hand side is at most linear in $\rho_{j_*}^\varepsilon$, while the left-hand side is quadratic. Hence $\rho_{j_*}^\varepsilon$ is bounded by a constant independent of ε . This proves the claim. $\square$

Boundedness of the effective Hamiltonian. The density bound controls the coefficients in the discrete eigenvalue problem and hence the stationary phase equation.

Proposition 3.2 (Uniform bound for the discrete effective Hamiltonian). *Assume that*

$$0 \leq \rho_j^\varepsilon \leq C_\rho, \quad j = 0, 1, \dots, N_x,$$

where C_ρ is independent of ε . Then the discrete effective Hamiltonian defined by (3.5) satisfies

$$|\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon)| \leq C_{\mathcal{H}}, \quad k = 1, \dots, N_\theta,$$

where $C_{\mathcal{H}}$ is independent of ε .

Proof. We use the trapezoidal weighted inner product

$$\langle \varphi, \psi \rangle_h = \Delta x \sum_{j=0}^{N_x} \omega_j \varphi_j \psi_j, \quad \omega_0 = \omega_{N_x} = \frac{1}{2}, \quad \omega_j = 1, \quad 1 \leq j \leq N_x - 1.$$

With the Neumann ghost values

$$\varphi_{-1} = \varphi_1, \quad \varphi_{N_x+1} = \varphi_{N_x-1},$$

the operator $-\delta_x^2$ is self-adjoint and nonnegative with respect to this weighted inner product. More precisely,

$$\langle -\delta_x^2 \varphi, \varphi \rangle_h = \Delta x \sum_{j=0}^{N_x-1} |\delta_x^+ \varphi_j|^2, \quad \delta_x^+ \varphi_j = \frac{\varphi_{j+1} - \varphi_j}{\Delta x}.$$

Hence the discrete eigenvalue problem is associated with the self-adjoint operator

$$-D_k \delta_x^2 - \text{diag}(K_j - \rho_j^\varepsilon)$$

in this weighted inner product. Its principal eigenvalue has the Rayleigh representation

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) = \min_{\varphi \neq 0} \frac{D_k \Delta x \sum_{j=0}^{N_x-1} |\delta_x^+ \varphi_j|^2 - \Delta x \sum_{j=0}^{N_x} \omega_j (K_j - \rho_j^\varepsilon) \varphi_j^2}{\Delta x \sum_{j=0}^{N_x} \omega_j \varphi_j^2}.$$

On the one hand, since $0 \leq K_j \leq K_{\max}$ and $\rho_j^\varepsilon \geq 0$, the nonnegative gradient term gives

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \geq -K_{\max}.$$

On the other hand, testing the quotient with the constant vector $\varphi_j \equiv 1$ gives

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \leq \frac{-\Delta x \sum_{j=0}^{N_x} \omega_j (K_j - \rho_j^\varepsilon)}{\Delta x \sum_{j=0}^{N_x} \omega_j} \leq C_\rho,$$

because $K_j \geq 0$ and $\rho_j^\varepsilon \leq C_\rho$. Therefore

$$|\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon)| \leq \max\{K_{\max}, C_\rho\} =: C_{\mathcal{H}},$$

which proves the claim. □

For the stationary phase equation, the Godunov numerical Hamiltonian further gives a fixed-grid bound for the phase. To be more specific, suppose that

$$\widehat{H}_G(\delta_\theta^- u_k^\varepsilon, \delta_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon), \quad (3.17)$$

where $\widehat{H}_G$ is defined in (A.1). Summing the stationary phase equation over the periodic trait grid eliminates $\delta_\theta^2 u^\varepsilon$. Therefore

$$\Delta\theta \sum_{k=1}^{N_\theta} |\delta_\theta^+ u_k^\varepsilon|^2 = -\Delta\theta \sum_{k=1}^{N_\theta} \widehat{H}_G(\delta_\theta^- u_k^\varepsilon, \delta_\theta^+ u_k^\varepsilon) = \Delta\theta \sum_{k=1}^{N_\theta} \mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \leq C_{\mathcal{H}}.$$

Thus

$$\max_k |\delta_\theta^+ u_k^\varepsilon| \leq \frac{\sqrt{C_{\mathcal{H}}}}{\sqrt{\Delta\theta}}. \quad (3.18)$$

Formal asymptotic-preserving trait-selection limit. We now pass formally to the fixed-grid rare-mutation limit. The density bound gives compactness of the discrete spatial marginal, and the Godunov phase estimate gives compactness of the normalized phase. Passing to the limit in the stationary phase equation yields a discrete constrained Hamilton–Jacobi structure.

Proposition 3.3 (Formal asymptotic-preserving structure with Godunov Hamiltonian). *Assume that the stationary semi-discrete states satisfy Assumption 3.1, and normalize the phase by*

$$\max_k u_k^\varepsilon = 0.$$

Assume also that the stationary phase equation is discretized with $\widehat{H}_G$ defined in (A.1). Then, up to a subsequence on the fixed spatial and trait grids,

$$\boldsymbol{\rho}^\varepsilon \rightarrow \boldsymbol{\rho}^0, \quad \mathbf{u}^\varepsilon \rightarrow \mathbf{u}^0.$$

Moreover, the limiting phase satisfies the discrete constrained Hamilton–Jacobi system

$$\widehat{H}_G(\delta_\theta^- u_k^0, \delta_\theta^+ u_k^0) = -\mathcal{H}_k(\boldsymbol{\rho}^0), \quad \max_k u_k^0 = 0.$$

If further $\mathcal{H}_k(\boldsymbol{\rho}^0)$ has a unique zero at k_m on the fixed trait grid, then

$$k^* = k_m,$$

where $k^ = \arg \max_k u_k^0$, meaning that the selected grid trait is exactly θ_{k_m} .*

Proof. Proposition 3.1 gives a uniform bound on $\boldsymbol{\rho}^\varepsilon$. Since the spatial grid is fixed, a subsequence converges to some $\boldsymbol{\rho}^0$. The preceding Godunov phase estimate and the normalization give a uniform bound on $\mathbf{u}^\varepsilon$ on the fixed trait grid. Hence, up to a further subsequence,

$$\mathbf{u}^\varepsilon \rightarrow \mathbf{u}^0.$$

Passing to the limit in the stationary phase equation gives the limiting discrete constrained Hamilton–Jacobi system, because $\varepsilon \delta_\theta^2 u_k^\varepsilon \rightarrow 0$ on the fixed trait grid and

$$\mathcal{H}_k(\boldsymbol{\rho}^\varepsilon) \rightarrow \mathcal{H}_k(\boldsymbol{\rho}^0).$$

The normalization also passes to the limit.

It remains to identify the selected grid trait. Let k_* be a maximizer of u^0 . Then

$$\delta_\theta^- u_{k_*}^0 \geq 0, \quad \delta_\theta^+ u_{k_*}^0 \leq 0.$$

By the definition of the Godunov Hamiltonian,

$$\widehat{H}_G(\delta_\theta^- u_{k_*}^0, \delta_\theta^+ u_{k_*}^0) = 0.$$

The limiting phase equation therefore gives

$$\mathcal{H}_{k_*}(\rho^0) = 0.$$

If $\mathcal{H}_k(\rho^0)$ has a unique zero at k_m , then every maximizer k_* of u^0 must satisfy $k_* = k_m$. $\square$

The fixed-grid asymptotic-preserving structure obtained above is the discrete analogue of the continuous rare-mutation structure summarized in Theorem 2.1. The discrete density estimate in Proposition 3.1 plays the role of the continuous bound on the spatial marginal. The discrete eigenvalue problem (3.5) defines the numerical effective Hamiltonian, corresponding to the continuous principal eigenvalue problem. By Proposition 3.3, the limiting discrete Hamilton–Jacobi system identifies the selected grid trait through the maximizer of the phase, in the same way that the continuous constrained Hamilton–Jacobi equation identifies the selected trait in the rare-mutation limit. Thus the semi-discrete WKB scheme preserves the essential selection structure on fixed spatial and trait grids.

The preceding argument is a stationary fixed-grid asymptotic-preserving structure. In the computations, the time-dependent WKB system is used as a relaxation dynamics toward such stationary states. Thus the semi-discrete analysis identifies the limiting stationary structure, while the numerical algorithm evolves the WKB variables toward this discrete eigenvalue and constrained Hamilton–Jacobi structure.

4 Numerical experiments

This section presents the fully discrete implementation and numerical experiments for the WKB formulation. We focus on its behavior in the small-mutation regime and compare it with direct density discretizations to demonstrate the advantage of the WKB formulation.

4.1 Fully discrete implementation and dual trait-grid implementation

We first describe the time discretization used in the numerical experiments. For comparison, the original density equation is discretized by the linear implicit update

$$\varepsilon \frac{n_{j,k}^{l+1} - n_{j,k}^l}{\Delta t} - D_k \delta_x^2 n_{j,k}^{l+1} - \varepsilon^2 \delta_\theta^2 n_{j,k}^{l+1} = n_{j,k}^{l+1} (K_j - \rho_j^l), \quad \rho_j^l = \Delta \theta \sum_{k=1}^{N_\theta} n_{j,k}^l. \quad (4.1)$$

For the WKB formulation, the phase variable u is advanced by

$$\frac{u_k^{l+1} - u_k^l}{\Delta t} + \widehat{H}(\delta_\theta^- u_k^l, \delta_\theta^+ u_k^l) - \varepsilon \delta_\theta^2 u_k^{l+1} = -H_k^l, \quad (4.2)$$

where H_k^l is determined by the discrete principal eigenvalue problem (3.5) with ρ_j^l reconstructed via (3.4). With u^{l+1} fixed, the amplitude W is advanced by the linear implicit update

$$\begin{aligned} \varepsilon \frac{W_{j,k}^{l+1} - W_{j,k}^l}{\Delta t} - D_k \delta_x^2 W_{j,k}^{l+1} - \varepsilon^2 \delta_\theta^2 W_{j,k}^{l+1} + 2\varepsilon D_\theta \widehat{F}(W^{l+1}, u^{l+1})_{j,k} \\ = W_{j,k}^{l+1} \left(K_j - \rho_j^l + H_k^l - 2\varepsilon \delta_\theta^2 u_k^{l+1} \right). \end{aligned} \quad (4.3)$$

Since u^{l+1} is already known, the amplitude update is also linear.

Large time-step relaxation. The time-dependent problem is used in this work as a relaxation dynamics toward the stationary selection state. In the small-mutation regime, the amplitude equation contains a fast spatial relaxation. Indeed, for frozen ρ^l and H_k^l , the leading x -dependent part of the amplitude equation is governed by the shifted operator

$$A_k(\rho^l) - H_k^l I, \quad A_k(\rho^l) = -D_k \delta_x^2 - \text{diag}(K - \rho^l),$$

whose null space is generated by the positive principal eigenvector associated with the discrete effective Hamiltonian. If $\mu_{\ell,k}^l > 0$ denotes a nonzero eigenvalue of $A_k(\rho^l) - H_k^l I$, where ℓ indexes the non-principal spatial modes, then the implicit treatment in (4.3) damps this mode by a factor of the form

$$\left(1 + \frac{\Delta t}{\varepsilon} \mu_{\ell,k}^l \right)^{-1}.$$

Therefore, when ε is very small, the WKB relaxation can use time steps much larger than ε to drive the amplitude rapidly toward the current principal-eigenvector profile. This is a practical advantage of the WKB relaxation solver.

Mass correction. In the time-dependent computations, we also use a scalar mass correction during the relaxation process. Integrating the density equation over x and θ , and using the homogeneous Neumann boundary condition in x and the periodic boundary condition in θ , gives

$$\varepsilon \frac{d}{dt} M_\varepsilon(t) = \int_\Omega \rho_\varepsilon(t, x) (K(x) - \rho_\varepsilon(t, x)) dx, \quad M_\varepsilon(t) = \int_\Omega \int_{\mathbb{T}} n_\varepsilon(t, x, \theta) d\theta dx.$$

After each time step, the provisional density is rescaled by a positive scalar so that the discrete total mass is consistent with the above mass evolution equation. In the computations below, this scalar factor is obtained using a trapezoidal rule in time. For the direct density solver, this amounts to rescaling n^{l+1} . For the WKB solver, we rescale the amplitude W^{l+1} , while the phase u^{l+1} is left unchanged. This correction is used only to enforce this global mass relation during the relaxation and does not change the location of the selected trait.

Dual trait-grid implementation. To improve computational efficiency, we introduce the dual trait-grid implementation. The phase variable u and the amplitude variable W are represented on different trait grids. Let Θ_u be the fine phase grid and Θ_W the coarser amplitude grid. When $\Theta_u = \Theta_W$, the method reduces to the standard single-grid WKB implementation. When Θ_W is coarser, the amplitude W is stored and evolved on fewer trait nodes, while the phase u still resolves the selected trait on the finer one-dimensional grid.

The coupling between the two trait grids is handled by periodic interpolation in the trait variable. Since u depends only on the trait variable, refining Θ_u is relatively inexpensive. By

contrast, W depends on both space and trait. In d spatial dimensions, the number of unknowns scales as

$$O\left(N_x^d N_\theta^{(W)} + N_\theta^{(u)}\right),$$

up to the cost of the spatial solvers. When $N_\theta^{(W)} \ll N_\theta^{(u)}$, the dual-grid strategy reduces the dominant memory and CPU cost while retaining fine trait resolution for the selection process.

4.2 Numerical experiments in 1D

Unless otherwise stated, the one-dimensional tests use

$$\Omega = [0, 1], \quad \mathbb{T} = (0, 1],$$

with homogeneous Neumann boundary conditions in the spatial variable and periodic boundary conditions in the trait variable. The carrying capacity and the dispersal rate are chosen as

$$K(x) = K_0 - K_1 \cos(2\pi x), \quad D(\theta) = D_{\min} + D_1(1 - \cos(2\pi(\theta - \theta_m))), \quad (4.4)$$

respectively, with $K_0 = 1$, $K_1 = 0.5$, $D_{\min} = 0.2$, $D_1 = 0.4$, $\theta_m = 0.35$. Thus D has a unique minimum at θ_m , which is the expected selected trait in the rare-mutation limit. The default initial data are chosen as

$$u_k^0 = 0, \quad W_{j,k}^0 = n_{j,k}^0 = 1. \quad (4.5)$$

Other initial data are specified in the individual tests when needed.

4.2.1 Numerical verification of Assumption 3.1

We first examine the one-sided weighted remainder condition used in the stationary asymptotic-preserving argument. For each computation, after the solution has reached a numerical steady state, we record the pairs

$$\left(\rho_j^\varepsilon, \mathcal{R}_j^{\varepsilon,D}\right), \quad j = 0, 1, \dots, N_x,$$

where $\mathcal{R}_j^{\varepsilon,D}$ is the weighted remainder (3.12).

Figure 1 shows representative scatter plots of $\mathcal{R}_j^{\varepsilon,D}$ against ρ_j^ε for several values of ε and for different dispersal functions $D(\theta)$. In the displayed tests, the recorded weighted remainders $\mathcal{R}_j^{\varepsilon,D}$ are negative. This behavior is consistent with the continuous expression (3.13). Indeed, for small ε , the density is concentrated near the selected trait, where $D(\theta)$ attains its minimum and, for the choices of D considered here, $1/D(\theta)$ has negative curvature. Hence the dominant contribution in (3.13) is expected to be nonpositive. Moreover, the magnitudes of the recorded remainders remain small relative to ρ_j^ε , which is also consistent with the prefactor ε^2 in (3.13). This provides supporting numerical evidence for Assumption 3.1 and for the stationary asymptotic-preserving argument of the WKB scheme.

4.2.2 Consistency between the direct density solver and the WKB formulation

We first perform a same-grid consistency check at finite ε . The direct density solver and the WKB solver are run on the same spatial and trait grids. The WKB density is reconstructed in the same way as in (3.3), and the corresponding spatial marginal $\rho_\varepsilon(t, x)$ and trait marginal $P_\varepsilon(t, \theta)$, both defined in (1.2), are compared with those obtained from the direct density computation.

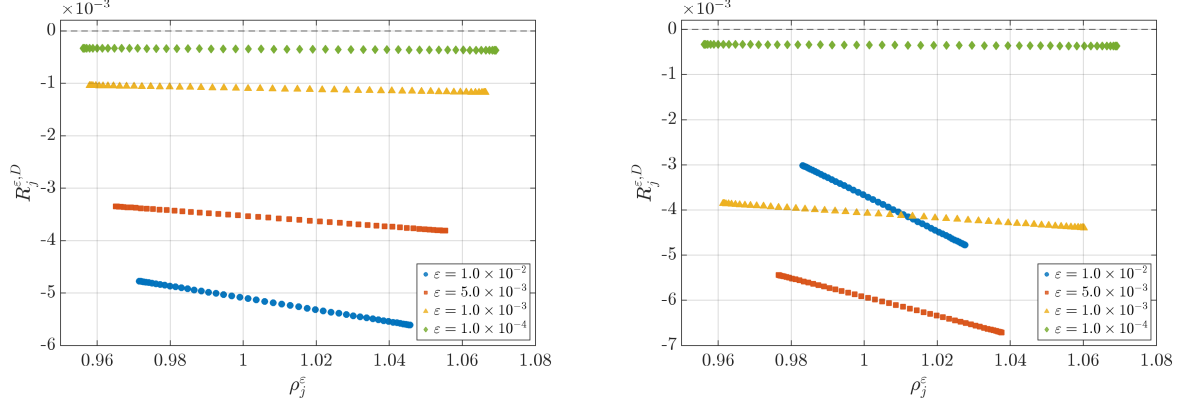


Figure 1: Numerical verification of Assumption 3.1. Left: $D(\theta) = 0.2 + 0.4(1 - \cos(2\pi(\theta - \theta_m)))$. Right: $D(\theta) = 0.2 + 0.2(1 - \cos(2\pi(\theta - \theta_m))) + 0.4(1 - \cos(6\pi(\theta - \theta_m)))$. The dashed horizontal line indicates zero.

Figure 2 shows the comparison at $t = 10$ with $\varepsilon = 10^{-3}$, $N_x = 100$, $N_\theta = 1024$ and $\Delta t = 10^{-3}$. The two solutions are almost indistinguishable for both the spatial marginal $\rho_\varepsilon(t = 10, x)$ and the trait marginal $P_\varepsilon(t = 10, \theta)$. This indicates that, when the trait grid is sufficiently fine, the WKB formulation is consistent with the direct density formulation at the density level.

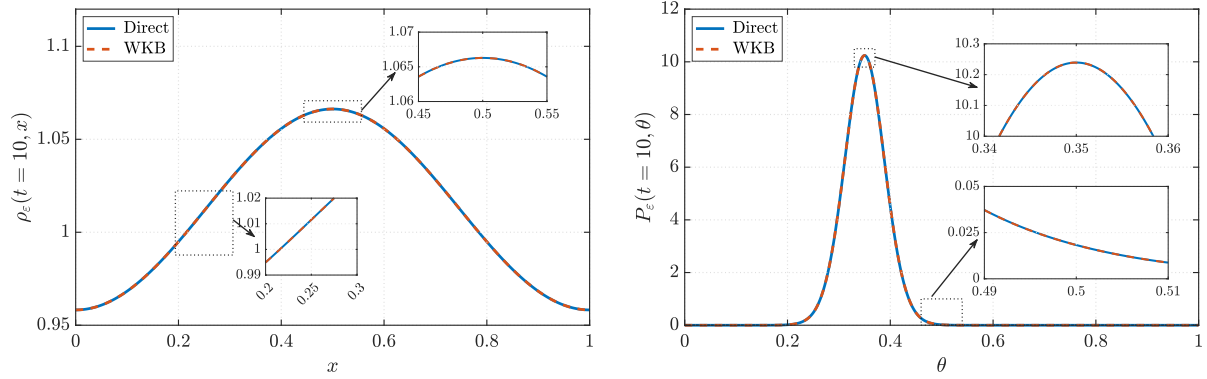


Figure 2: Same-grid comparison between the direct density solver and the WKB formulation. Left: spatial marginal $\rho_\varepsilon(t = 10, x)$. Right: trait marginal $P_\varepsilon(t = 10, \theta)$. Parameters are $\varepsilon = 10^{-3}$, $t = 10$, $N_x = 100$, $N_\theta = 1024$ and $\Delta t = 10^{-3}$.

Table 1 further reports the same-grid errors as the trait grid is refined. The errors decrease steadily with N_θ . After the trait grid is moderately refined, the error is reduced by roughly a factor of four whenever N_θ is doubled, which is consistent with the second-order trait discretization. This test confirms that the WKB phase-amplitude formulation does not introduce a density-level inconsistency when the direct density profile is sufficiently resolved.

4.2.3 Long-time concentration in the trait direction

Here we illustrate the concentration process in the trait direction. According to the rare-mutation asymptotics, the trait marginal $P_\varepsilon(t, \theta)$ is expected to concentrate near the selected trait as time evolves, and the concentration becomes more pronounced as ε decreases.

Table 1: Same-grid consistency between the direct density solver and the WKB reconstruction at $t = 10$ with $\varepsilon = 10^{-3}$, $N_x = 100$ and $\Delta t = 10^{-3}$. Here the errors are measured in the discrete ℓ^2 - and ℓ^∞ -norms over the space-trait grid.

N_θ	16	32	64	128	256	512	1024
$\ n_{direct} - n_{WKB}\ _2$	1.74E-1	3.84E-2	8.70E-3	2.13E-3	5.30E-4	1.32E-4	3.30E-5
$\ n_{direct} - n_{WKB}\ _\infty$	1.42E-1	4.59E-2	1.03E-2	2.63E-3	6.54E-4	1.63E-4	4.08E-5

Figure 3 shows this behavior. In the left panel, ε is fixed and the trait marginal is plotted at several time levels. Starting from a broad initial profile, the peak moves toward the expected selected trait $\theta_m = 0.35$, and the solution approaches a sharply localized steady profile. In particular, the profiles at the later times are almost indistinguishable, indicating that a numerical steady state has been reached. In the right panel, the final-time marginals are compared for different values of ε . The peak becomes narrower and higher as ε decreases, showing the concentration of the trait distribution toward θ_m . These results are consistent with the continuous rare-mutation structure and demonstrate that the WKB formulation captures the emergence of the selected trait.

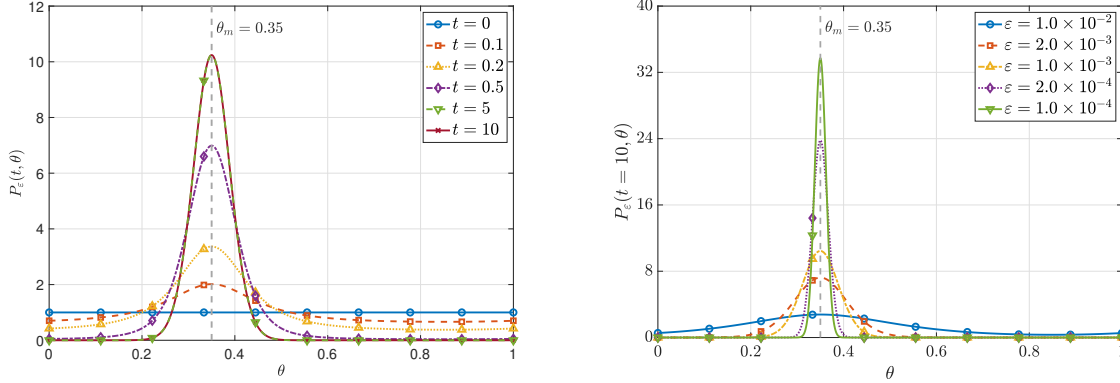


Figure 3: Long-time concentration in the trait direction. Left: evolution of $P_\varepsilon(t, \theta)$ for fixed $\varepsilon = 10^{-3}$, $N_x = 100$, $N_\theta = 128$, and $\Delta t = 10^{-3}$. Right: final-time trait marginals for different values of ε . The vertical dashed line marks the expected selected trait $\theta_m = 0.35$.

4.2.4 WKB versus direct density discretization on coarse trait grids

We next compare the WKB formulation with direct density discretizations on coarse trait grids, aiming to illustrate the main numerical advantage of the WKB representation. In the small-mutation regime, the direct density becomes sharply concentrated in the trait direction, and a coarse trait grid may fail to resolve the location and shape of the concentration. By contrast, in the WKB formulation the selected trait is encoded in the phase variable, which remains much smoother than the density.

For both the plots and the peak diagnostics in this subsection, we use the same log-based reconstruction of the trait marginal. For a direct density computation, we compute

$$\log P_\varepsilon(t, \theta) = \log \left(\int_{\Omega} n_\varepsilon(t, x, \theta) dx \right). \quad (4.6)$$

Similarly, for a WKB computation, we compute

$$\log P_\varepsilon(t, \theta) = \frac{u_\varepsilon(t, \theta)}{\varepsilon} + \log \left(\int_{\Omega} W_\varepsilon(t, x, \theta) dx \right). \quad (4.7)$$

This logarithmic evaluation avoids numerical underflow or overflow when ε is small. When a spatial marginal or a total mass is computed from the WKB representation, we use a similar Laplace-type asymptotic approximation to evaluate sums involving $\exp(u_\varepsilon(t, \theta)/\varepsilon)$ [25, 26]. The interpolation is then applied to $\log P_\varepsilon(t, \theta)$, rather than to $P_\varepsilon(t, \theta)$ itself.

Figure 4 shows the comparison for decreasing values of ε . The direct density solutions are plotted for several trait resolutions, while the WKB result is shown as the reference WKB computation on the same physical problem. For moderate ε , both approaches give comparable results once the trait grid is sufficiently refined. As ε decreases, however, the direct density solver becomes much more sensitive to the trait resolution. In particular, coarse direct grids give visibly shifted or broadened trait marginals. The WKB formulation remains well aligned with the selected trait because the phase variable resolves the concentration location without directly resolving the full density profile.

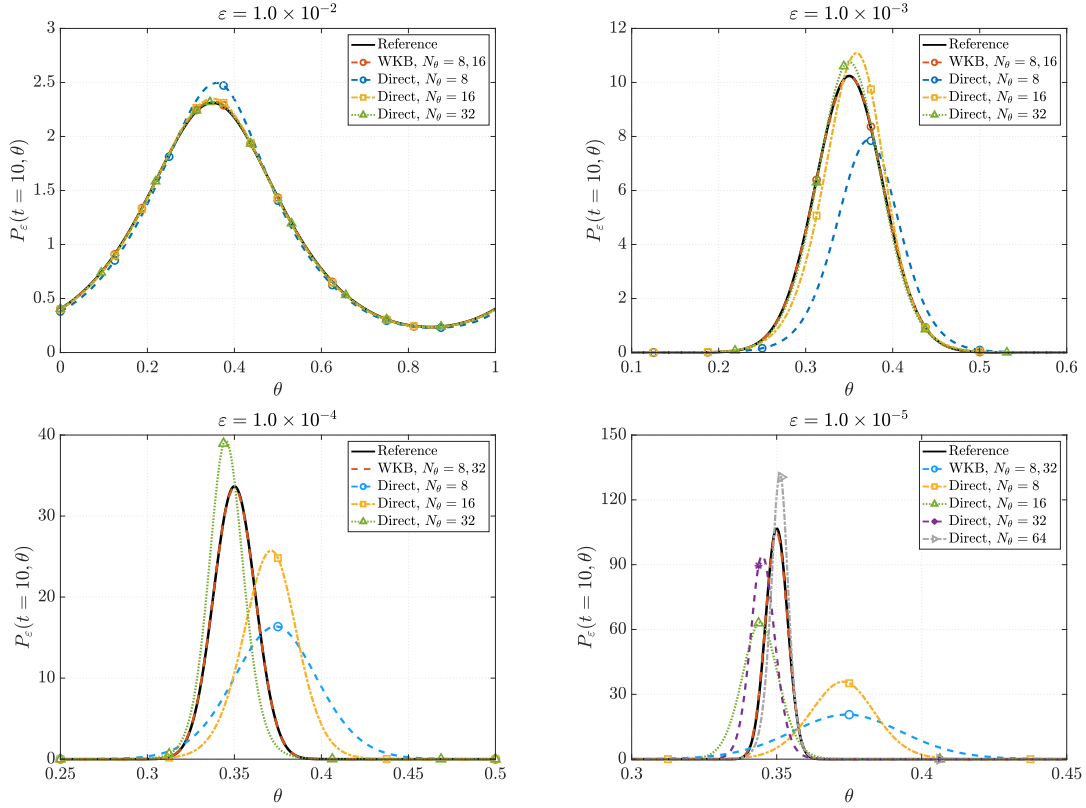


Figure 4: Comparison between the WKB formulation and direct density discretizations on coarse trait grids. The four panels correspond to $\varepsilon = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$, respectively. All trait marginals are reconstructed using the same log-based postprocessing described in the text. The direct density solver requires increasingly fine trait grids to resolve the concentrated marginal, whereas the WKB formulation continues to capture the selected trait through the phase variable.

This experiment shows that the benefit of the WKB formulation becomes more pronounced as ε decreases. The result is consistent with the asymptotic picture that the density-level trait

marginal develops a narrow concentration, while the WKB phase remains a more regular quantity from which the selected trait can be identified on a fixed trait grid.

To quantify this comparison, we extract several diagnostics from the reconstructed trait marginal $P_\varepsilon(t, \theta)$, including the peak-location error e_θ , the peak-height error e_{PH} , and the peak-width error e_{PW} , where the peak width is measured by the full width at half maximum. Here we remark that, in the small ε regime, post-processing techniques such as a local quadratic fit of $\log P_\varepsilon(t, \theta)$ near the peak may be applied to further improve the performance of the WKB formulation.

We next examine two sources of computational efficiency gained by the WKB formulation. First, the WKB representation reduces the dominant number of space–trait degrees of freedom needed to resolve the concentrated marginal. We compare direct density discretizations with $N_\theta^{(n)}$, single-grid WKB computations with $N_\theta^{(u)} = N_\theta^{(W)}$, and dual-grid WKB computations with $N_\theta^{(W)} < N_\theta^{(u)}$. Here the superscript specifies which variable the trait grid is associated with. The effective number of unknowns is measured by

$$N_{\text{dof}}^{\text{dir}} = (N_x + 1) \cdot N_\theta^{(n)}$$

for the direct density solver, and by

$$N_{\text{dof}}^{\text{WKB}} = (N_x + 1) \cdot N_\theta^{(W)} + N_\theta^{(u)}$$

for the WKB formulation. This count reflects the fact that the amplitude W carries all spatial degrees of freedom, whereas the phase u is only a one-dimensional trait variable. Using a single-grid WKB solution with $N_\theta = 1024$ as the reference, Figure 5 reports the errors in the selected trait and peak diagnostics against the effective number of degrees of freedom. The single-grid WKB results converge rapidly as the trait grid is refined. The dual-grid WKB results show that one can keep a relatively fine phase grid while using a much coarser amplitude grid, with only a mild loss of accuracy. In contrast, the direct density solver requires substantially more trait degrees of freedom to resolve the concentrated marginal. This confirms one main computational advantage of the WKB formulation, namely that the concentration location is resolved through the low-dimensional phase, rather than by evolving the full space–trait density on a fine trait grid.

Second, the WKB representation also allows a larger stable time step in the small- ε regime. In Figure 6, we compare the admissible time steps of the WKB and direct density solvers over a range of ε for a fixed trait resolution. The WKB formulation remains stable for larger time steps, especially when ε is small, whereas the direct density solver becomes more restrictive. This improvement is closely related to the asymptotic-preserving structure of the WKB scheme. In the rare-mutation regime, the direct solver has to evolve an exponentially concentrated density profile. The WKB scheme instead separates the leading exponential scale and preserves the limiting Hamilton–Jacobi structure at the discrete level. Hence its admissible time step is less constrained by the singular concentration scale of the original density variable.

Overall, the observed computational gain is not only due to the reduced effective number of unknowns, but also to the improved time-step robustness of the WKB reformulation in the rare-mutation regime.

4.3 Two-dimensional spatial experiment

We finally present a two-dimensional spatial experiment to illustrate that the WKB formulation can be applied beyond the one-dimensional spatial setting. The computational domain is $\Omega = [0, 1]^2$, and we choose the spatial growth rate and the dispersal function as

$$K(x, y) = K_0 + K_1 \cos(2\pi x) \cos(2\pi y), \quad D(\theta) = D_{\min} + D_1(1 - \cos(2\pi(\theta - \theta_m))).$$

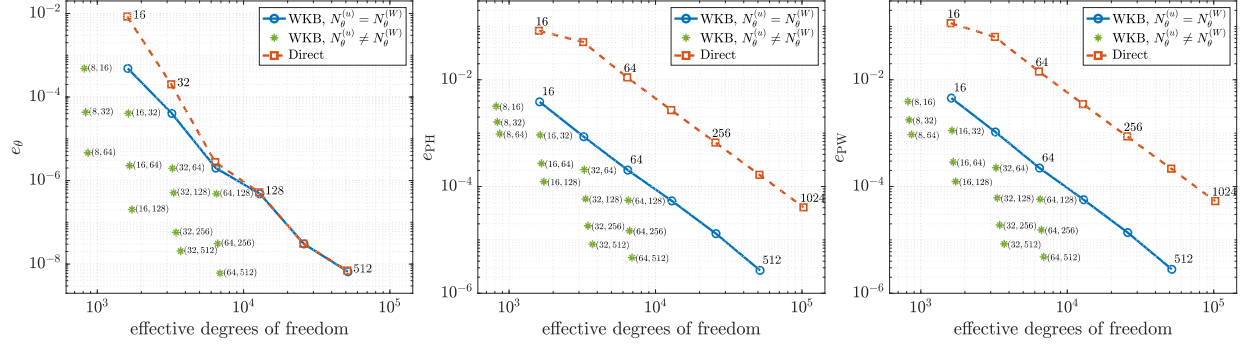


Figure 5: Efficiency comparison in terms of effective degrees of freedom. The direct density solver uses $N_{\text{dof}}^{\text{dir}} = (N_x + 1) \cdot N_\theta^{(n)}$, while the WKB formulation uses $N_{\text{dof}}^{\text{WKB}} = (N_x + 1) \cdot N_\theta^{(W)} + N_\theta^{(u)}$. The dual-grid WKB runs use a fine phase grid and a coarser amplitude grid. The results show that the WKB formulation can achieve comparable or better peak diagnostics with much fewer degrees of freedom.

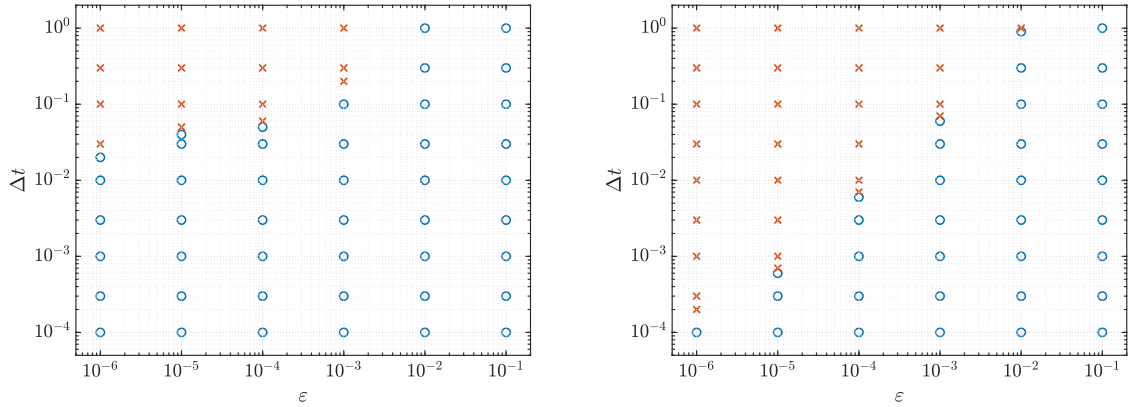


Figure 6: Time-step robustness for fixed trait resolution $N_\theta = 64$. Left: WKB solver. Right: direct density solver. Blue circles denote stable computations, whereas red crosses denote unstable computations. The WKB formulation permits larger admissible time steps in the small- ϵ regime.

In the computation, we take

$$K_0 = 1, \quad K_1 = 0.5, \quad D_{\min} = 0.2, \quad D_1 = 0.4, \quad \theta_m = 0.35.$$

Figure 7 shows the numerical result with $\varepsilon = 10^{-3}$, $N_x = N_y = 50$, and $N_\theta = 64$ at $t = 20$. The left panel displays the spatial marginal density $\rho_\varepsilon(t, x, y)$, which reflects the spatial heterogeneity induced by $K(x, y)$. The right panel shows the corresponding trait marginal $P_\varepsilon(t, \theta)$. The trait distribution is sharply localized near the expected selected trait $\theta_m = 0.35$, indicating that the WKB computation captures the same rare-mutation concentration mechanism in the two-dimensional spatial case.

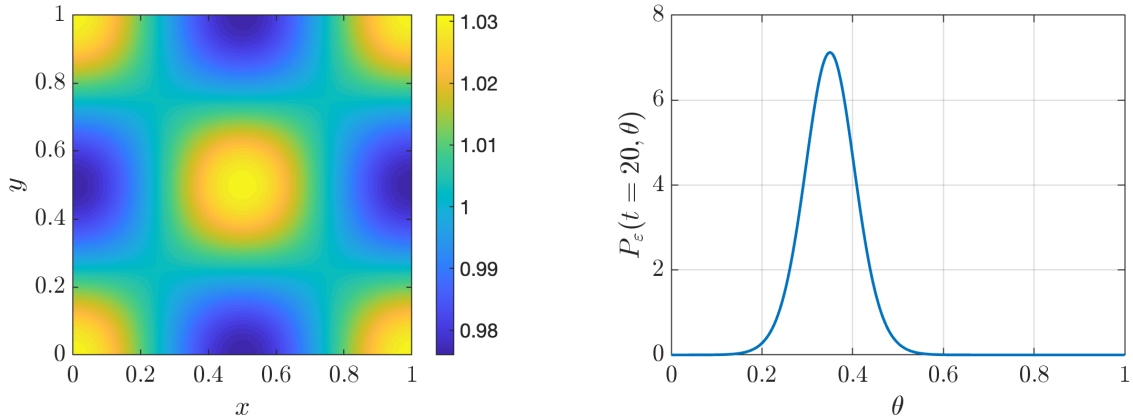


Figure 7: Two-dimensional spatial experiment with $\varepsilon = 10^{-3}$, $N_x = N_y = 50$, and $N_\theta = 64$ at $t = 20$. Left: spatial marginal density $\rho_\varepsilon(t, x, y)$. Right: trait marginal $P_\varepsilon(t, \theta)$.

5 Conclusion

In this paper, we developed a WKB-based fixed-grid method for a dispersal evolution model in the rare-mutation regime. We proved positivity preservation and established a stationary fixed-grid asymptotic-preserving structure under a one-sided weighted remainder condition. Numerical experiments show that the method accurately recovers the selected trait on coarse trait grids, reduces the dominant space-trait degrees of freedom through a dual-grid implementation, and allows larger stable time steps for small ε . A two-dimensional test further illustrates its ability to capture spatial heterogeneity and trait concentration beyond the one-dimensional setting.

A Numerical Hamiltonians and conservative fluxes

For the monotone theory in Section 3, the numerical Hamiltonian $\hat{H}(a, b)$ is assumed to be consistent with $-p^2$, nondecreasing in its first argument, and nonincreasing in its second argument. For any real number r , let

$$r_+ := \max\{r, 0\}, \quad r_- := \min\{r, 0\}.$$

One typical choice is the Godunov Hamiltonian

$$\hat{H}_G(a, b) = -(a_-)^2 - (b_+)^2. \tag{A.1}$$

In practice, high-order one-sided WENO reconstructions can be applied [13, 21, 24]. This is a numerical option for improving resolution of the phase profile.

The conservative flux in the equation of W can be chosen as

$$D_\theta \widehat{F}(W, u)_{j,k} = \frac{\widehat{F}_{j,k+1/2} - \widehat{F}_{j,k-1/2}}{\Delta\theta}, \quad a_{k+1/2} = -\frac{u_{k+1} - u_k}{\Delta\theta},$$

with the upwind flux

$$\widehat{F}_{j,k+1/2}^{\text{up}} = (a_{k+1/2})_+ W_{j,k} + (a_{k+1/2})_- W_{j,k+1}. \quad (\text{A.2})$$

B Remainder bound for a conservative density discretization

We give a simple situation in which Assumption 3.1 can be verified. Suppose that the reconstructed density satisfies the conservative semi-discrete density equation

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon = (K_j - \rho_j^\varepsilon) n_{j,k}^\varepsilon. \quad (\text{B.1})$$

Then periodic summation by parts gives

$$\mathcal{R}_j^{\varepsilon,D} = \varepsilon^2 \Delta\theta \sum_k n_{j,k}^\varepsilon \delta_\theta^2 \left(\frac{1}{D_k} \right).$$

In this case,

$$(R_j^{\varepsilon,D})_+ \leq \varepsilon^2 C_{D,h} \rho_j^\varepsilon, \quad C_{D,h} = \max_k \left| \delta_\theta^2 \left(\frac{1}{D_k} \right) \right|.$$

This verifies Assumption 3.1 with $C_{R,h} = C_{D,h}$ for $\varepsilon \in (0, 1]$ and $r_h = 0$.

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